

Industrial Substation & MCC Design Report

1. Project Overview

This project involved the design and analysis of a simplified industrial electrical distribution system for a food-processing and packaging facility.

The system was developed using EPLAN Education for electrical schematics and MCC diagrams, PowerWorld Simulator for power-flow analysis, and Google Sheets for electrical calculations, equipment sizing, operating-case comparisons, and graphs.

The design includes a 13.8 kV utility supply, a 750 kVA transformer, a 480 V main switchboard, two motor control centers, an auxiliary load section, six industrial motors, motor protection, VFD and DOL motor control, and four operating scenarios.

The purpose of the project was to combine industrial electrical design, motor control, equipment sizing, schematic development, and power-system simulation into one complete portfolio project.

2. Project Objectives

The main objectives of the project were to:

- Design a simplified industrial substation and low-voltage distribution system
 - Select an appropriate transformer rating
 - Develop a 480 V main switchboard and MCC distribution layout
 - Size main feeders and motor branch circuits
 - Design motor control and protection circuits
 - Represent both DOL and VFD-controlled motors
 - Create single-line and multi-line electrical drawings in EPLAN
 - Build a power-flow model in PowerWorld
 - Evaluate bus voltages and equipment loading
 - Compare system performance under multiple operating conditions
 - Identify the limiting equipment for future plant expansion
-

3. System Description

The plant is supplied from a 13.8 kV, 3-phase, 60 Hz utility source.

The utility supply feeds a 750 kVA transformer that steps the voltage down from 13.8 kV to 480 V.

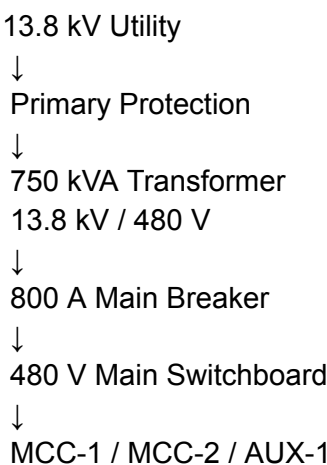
The transformer supplies a main 480 V switchboard, SWBD-1.

The main switchboard supplies three major downstream sections:

- MCC-1
- MCC-2
- AUX-1

MCC-1 and MCC-2 each supply multiple industrial motor loads, while AUX-1 represents additional plant auxiliary loads.

The overall system structure is:



4. Load Schedule

The plant contains six main motor loads and one equivalent auxiliary load.

Tag	Equipment	Control Method	Output Power
M-101	Process Pump	VFD	75 kW
M-102	Cooling Water Pump	DOL	45 kW

M-103	Main Conveyor	VFD	30 kW
M-201	Exhaust Fan	VFD	55 kW
M-202	Air Compressor	DOL	90 kW
M-203	Secondary Conveyor	DOL	22 kW
AUX-1	Auxiliary Loads	Equivalent Load	100 kVA

Motor input power, apparent power, reactive power, and current were calculated using the selected efficiency and power-factor assumptions for each motor.

The approximate total plant demand was calculated as:

- Real Power: 432.5 kW
- Reactive Power: 232.6 kvar
- Apparent Power: 491.1 kVA
- Plant Current: 590.7 A
- Plant Power Factor: approximately 0.88

5. Transformer Selection

A 750 kVA transformer was selected for the plant.

Transformer specifications:

- Rating: 750 kVA
- Primary Voltage: 13.8 kV
- Secondary Voltage: 480 V
- Phase: 3-phase
- Frequency: 60 Hz
- Assumed Impedance: 5.75%

The calculated initial plant demand was approximately 491.1 kVA.

Therefore, the initial transformer loading was:

$$491.1 \text{ kVA} / 750 \text{ kVA} \approx 65.5\%$$

The transformer full-load secondary current was approximately 902 A.

The 750 kVA transformer therefore provides suitable capacity for the normal plant load while also leaving additional capacity for moderate future growth.

6. Distribution and Feeder Design

The main distribution system was sized using the calculated plant and motor load currents.

The selected feeder ratings were:

Feeder	Destination	Selected Rating
FDR-MAIN	Transformer to SWBD-1	800 A
FDR-01	SWBD-1 to MCC-1	300 A
FDR-02	SWBD-1 to MCC-2	300 A
FDR-03	SWBD-1 to AUX-1	175 A

The selected bus ratings were:

- SWBD-1: 800 A
- MCC-1: 300 A
- MCC-2: 300 A
- AUX-1: 175 A

Motor branch ratings were selected as follows:

Motor	Control Method	Selected Rating
M-101	VFD	150 A
M-102	DOL	100 A
M-103	VFD	60 A
M-201	VFD	125 A
M-202	DOL	175 A
M-203	DOL	50 A

These ratings were used as simplified design assumptions for a portfolio project and are not intended to replace detailed construction-level electrical code calculations.

7. EPLAN Electrical Design

EPLAN Education was used to create the electrical documentation for the plant.

The EPLAN portion of the project included:

- Substation single-line diagram
- Main switchboard representation
- MCC-1 schematic
- MCC-2 schematic
- DOL motor starter circuits
- VFD motor branches
- Motor protection
- Start and stop control circuits
- Overload trip contacts
- Contactor coils
- Auxiliary seal-in contacts

Substation Single-Line

The single-line diagram represents the complete distribution path from the utility source to the downstream MCC and auxiliary loads.

The main sequence is:

Utility Source

↓

Primary Protection

↓

Transformer T1

↓

Main LV Breaker

↓

SWBD-1

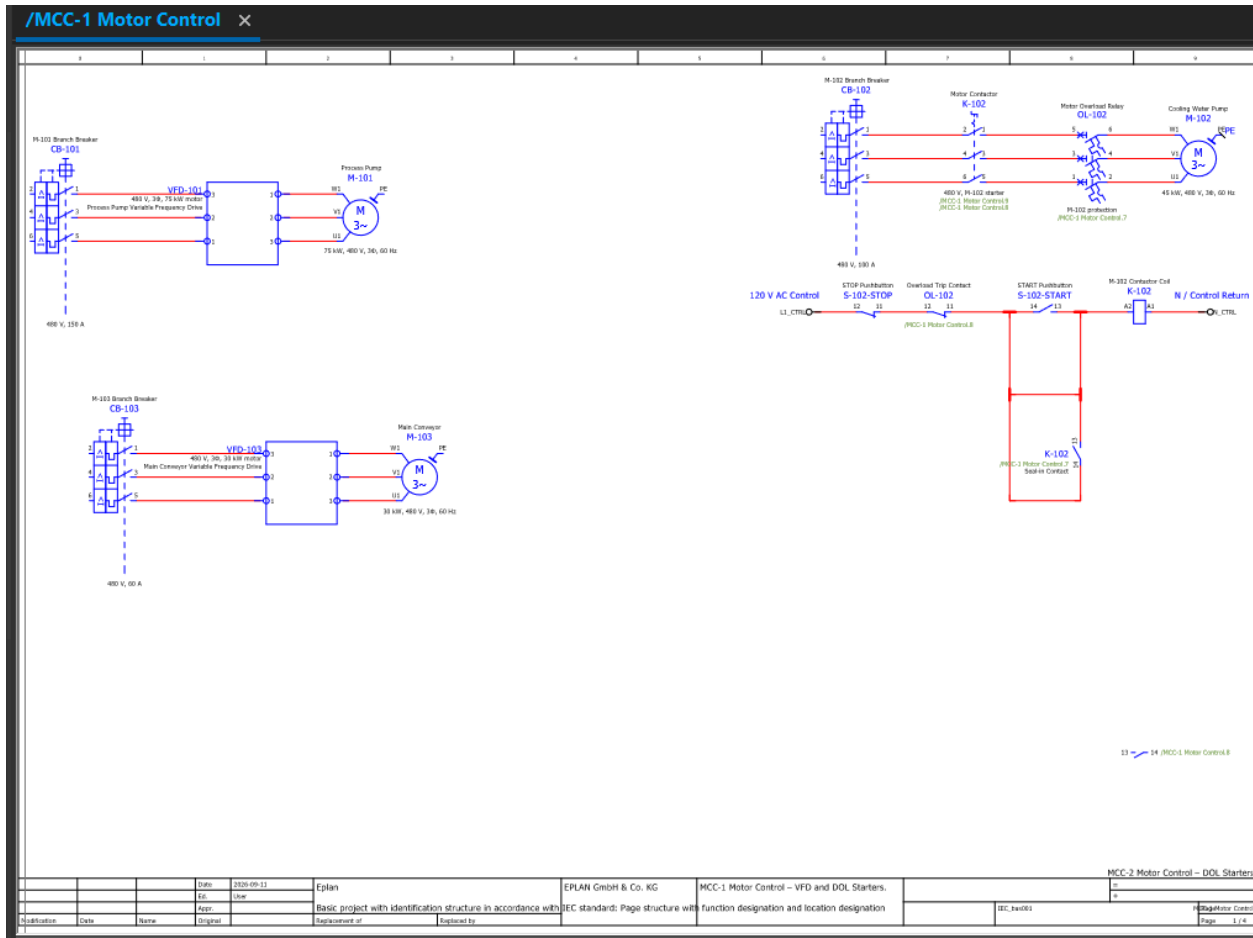
↓

MCC-1 / MCC-2 / AUX-1



MCC-1 includes:

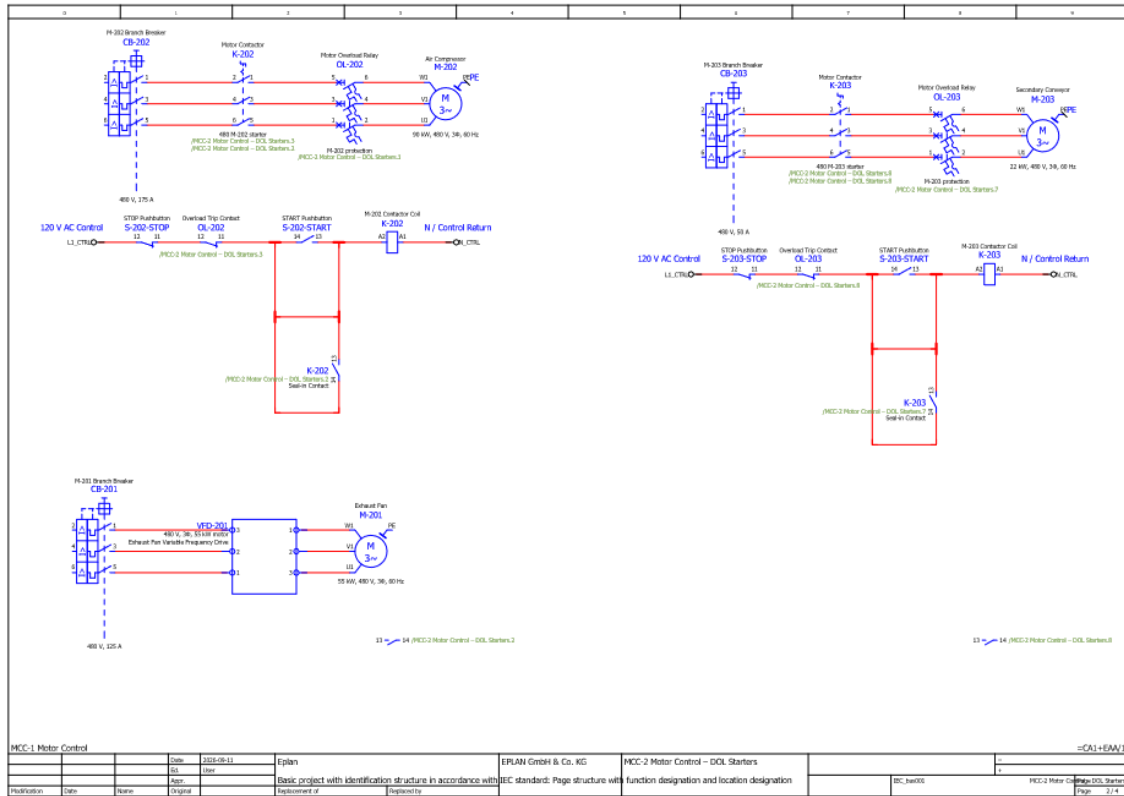
- M-101 Process Pump, 75 kW, VFD
- M-102 Cooling Water Pump, 45 kW, DOL
- M-103 Main Conveyor, 30 kW, VFD



MCC-2

MCC-2 includes:

- M-201 Exhaust Fan, 55 kW, VFD
- M-202 Air Compressor, 90 kW, DOL
- M-203 Secondary Conveyor, 22 kW, DOL



DOL Starter Design

The DOL starter power circuit includes:

Breaker



Contactor



Overload Relay



3-Phase Motor

The control circuit includes:

120 V AC Control



STOP Normally Closed



Overload Trip Normally Closed



START Normally Open



Contactor Coil



Control Return

A normally open auxiliary contact is connected in parallel with the START pushbutton to create the seal-in circuit.

VFD Motor Design

The VFD-controlled motor branches use the simplified structure:

Breaker



Variable Frequency Drive



3-Phase Motor

The VFD-controlled motors are:

- M-101 Process Pump
 - M-103 Main Conveyor
 - M-201 Exhaust Fan
-

8. PowerWorld Model

A simplified five-bus system was created in PowerWorld Simulator.

The buses were:

Bus	Description	Nominal Voltage
BUS-01	Utility	13.8 kV
BUS-02	Main	0.48 kV
BUS-03	MCC-1	0.48 kV
BUS-04	MCC-2	0.48 kV
BUS-05	AUX-1	0.48 kV

BUS-01 was configured as the system slack bus and included the utility generator.

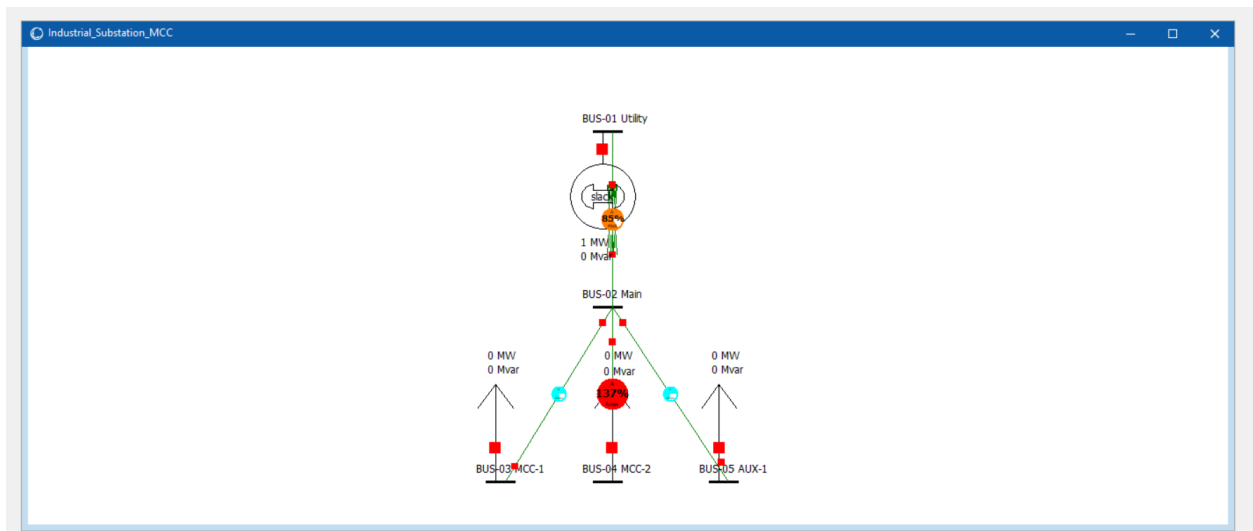
The model also included:

- Transformer T1
- Feeder from BUS-02 to BUS-03
- Feeder from BUS-02 to BUS-04
- Feeder from BUS-02 to BUS-05
- Aggregate MCC-1 load
- Aggregate MCC-2 load
- Auxiliary load
- Branch MVA limits

The power flow was solved using the Newton-Raphson method.

The completed model successfully validated with:

- 0 errors
- 0 warnings
- Successful power-flow convergence



9. Operating Cases

Four operating scenarios were evaluated.

Case 1 — Normal Operation

This case represents the normal plant operating condition.

Results:

- BUS-02 Voltage: 0.98126 pu
- BUS-03 Voltage: 0.98125 pu
- BUS-04 Voltage: 0.98125 pu
- BUS-05 Voltage: 0.98125 pu
- Transformer Loading: 66.7%
- FDR-01 Loading: 75.2%
- FDR-02 Loading: 81.9%
- FDR-03 Loading: 69.0%
- Utility Power: 0.43 MW
- Utility Reactive Power: 0.25 Mvar

The system remains within the selected equipment ratings under normal operating conditions.

Case 2 — Heavy Production

This case represents an approximately 10% increase in plant loading.

Results:

- BUS-02 Voltage: 0.97895 pu
- BUS-03 Voltage: 0.97894 pu
- BUS-04 Voltage: 0.97894 pu
- BUS-05 Voltage: 0.97894 pu
- Transformer Loading: 74.4%
- FDR-01 Loading: 82.7%
- FDR-02 Loading: 91.7%
- FDR-03 Loading: 77.1%
- Utility Power: 0.48 MW
- Utility Reactive Power: 0.28 Mvar

The additional plant demand causes a small reduction in bus voltage and increases equipment loading. However, the transformer and all three feeders remain below their selected limits.

Case 3 — M-202 Air Compressor Offline

This scenario represents the 90 kW M-202 air compressor being removed from operation.

Results:

- BUS-02 Voltage: 0.98567 pu
- BUS-03 Voltage: 0.98567 pu
- BUS-04 Voltage: 0.98567 pu
- BUS-05 Voltage: 0.98567 pu

- Transformer Loading: 50.8%
- FDR-01 Loading: 73.7%
- FDR-02 Loading: 37.9%
- FDR-03 Loading: 67.9%
- Utility Power: 0.33 MW
- Utility Reactive Power: 0.19 Mvar

Removing M-202 significantly reduces plant demand and MCC-2 feeder loading.

The MCC-2 feeder decreases from 81.9% loading during normal operation to 37.9%.

Case 4 — Future Expansion

This case evaluates a future expansion involving an additional 110 kW motor connected to MCC-2.

Results:

- BUS-02 Voltage: 0.97636 pu
- BUS-03 Voltage: 0.97636 pu
- BUS-04 Voltage: 0.97636 pu
- BUS-05 Voltage: 0.97636 pu
- Transformer Loading: 84.5%
- FDR-01 Loading: 74.3%
- FDR-02 Loading: 134.1%
- FDR-03 Loading: 69.0%
- Utility Power: 0.55 MW
- Utility Reactive Power: 0.32 Mvar

This case produced the most important engineering result in the project.

Although the main transformer remains below its rating at 84.5% loading, the MCC-2 feeder reaches 134.1% loading.

This indicates that the transformer itself has sufficient capacity for the proposed expansion, but the existing MCC-2 feeder does not.

10. Results Comparison

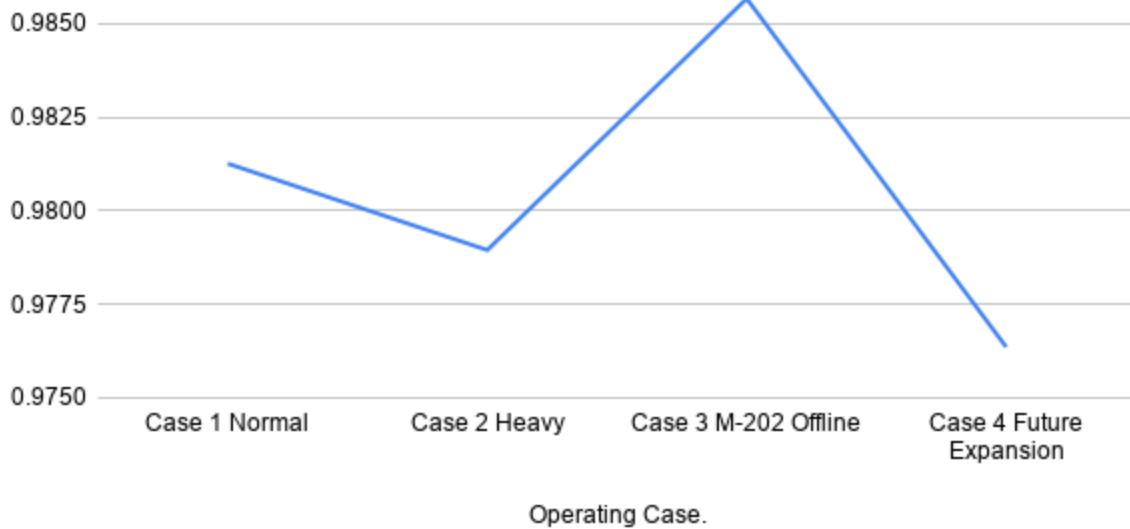
Metric	Case 1 Normal	Case 2 Heavy	Case 3 M-202 Offline	Case 4 Expansion
--------	------------------	-----------------	-------------------------	---------------------

BUS-02 Voltage (pu)	0.98126	0.97895	0.98567	0.97636
BUS-03 Voltage (pu)	0.98125	0.97894	0.98567	0.97636
BUS-04 Voltage (pu)	0.98125	0.97894	0.98567	0.97636
BUS-05 Voltage (pu)	0.98125	0.97894	0.98567	0.97636
T1 Loading (%)	66.7	74.4	50.8	84.5
FDR-01 Loading (%)	75.2	82.7	73.7	74.3
FDR-02 Loading (%)	81.9	91.7	37.9	134.1
FDR-03 Loading (%)	69.0	77.1	67.9	69.0
Utility MW	0.43	0.48	0.33	0.55
Utility Mvar	0.25	0.28	0.19	0.32

11. Results and Analysis

Bus Voltage Comparison

Main Bus Voltage Across Operating Cases

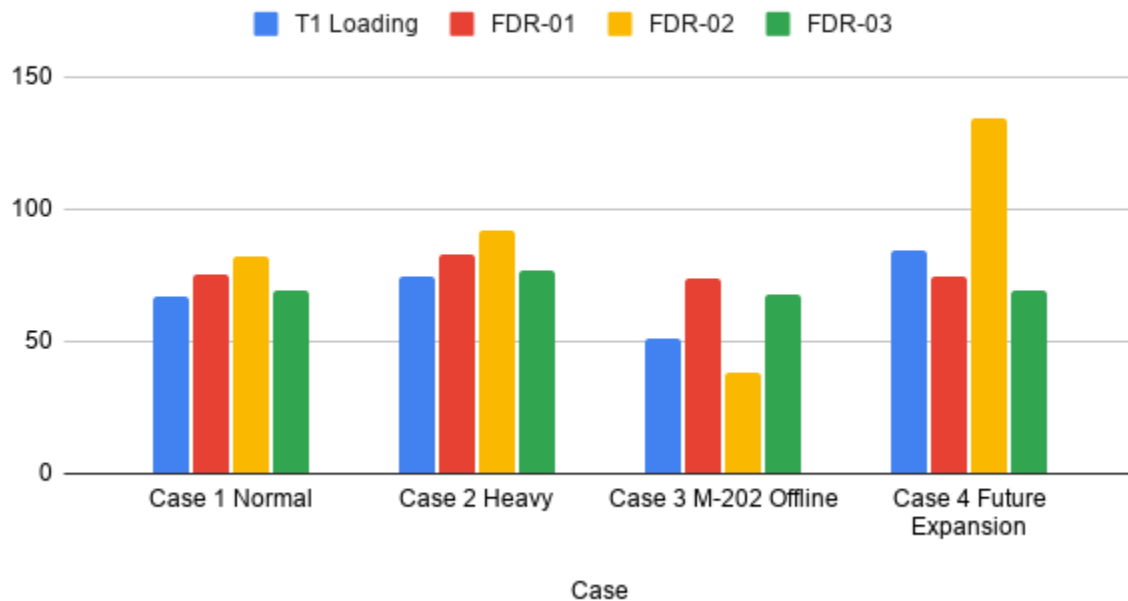


Observation: Bus voltage remained close to nominal for all operating conditions. The highest voltage occurred with M-202 offline, while the lowest voltage occurred during the future expansion case due to the increased system loading.

The voltage variation across the operating cases was relatively small. This indicates that the system is able to maintain acceptable voltage levels even under the heavier loading scenarios considered in this study.

Transformer and Feeder Loading

Transformer and Feeder Loading Across Operating Cases

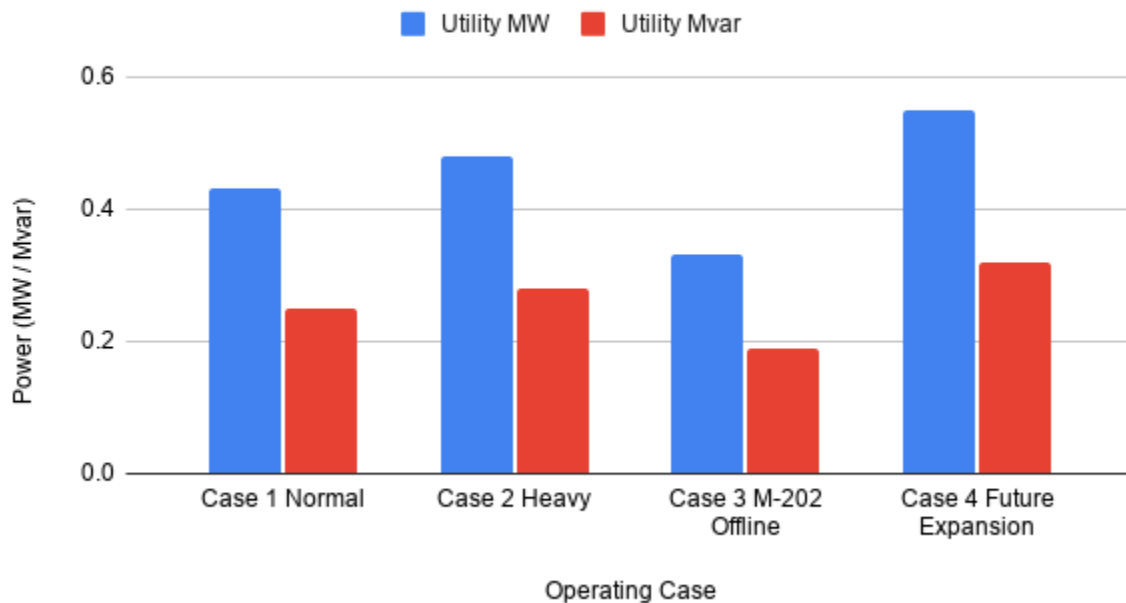


Observation: Equipment loading varies significantly across operating conditions. The most critical result occurs in Case 4, where FDR-02 reaches 134.1% loading and exceeds its rated capacity. The transformer remains below its rating at 84.5%, so the main limitation for future expansion is the MCC-2 feeder.

The results show that the transformer is not the limiting component of the system. Instead, the MCC-2 feeder becomes the first major bottleneck as additional load is added.

Utility Power Demand

Utility Power Demand Across Operating Cases



Observation: Utility demand is lowest when M-202 is offline and highest during the future expansion case. Real power demand increases from 0.43 MW under normal operation to 0.55 MW in the future expansion case, while reactive power increases from 0.25 Mvar to 0.32 Mvar.

The utility demand follows the expected pattern across all operating scenarios. Removing a large motor reduces both real and reactive power demand, while adding future load increases both quantities.

12. Key Engineering Findings

The completed analysis produced several important engineering findings.

First, the system performs acceptably under normal operating conditions. The transformer and all distribution feeders remain below their selected limits.

Second, the heavy-production case increases loading but does not create an overload. The most heavily loaded feeder in this scenario is FDR-02 at 91.7%.

Third, removing the M-202 air compressor has a significant effect on the MCC-2 system. The MCC-2 feeder loading drops from 81.9% to 37.9%, demonstrating the large contribution of the 90 kW compressor to the plant demand.

Fourth, the future expansion scenario identifies a major system limitation. Adding a future 110 kW motor causes FDR-02 to reach 134.1% of its selected limit.

Finally, the transformer does not become overloaded during the future expansion case. Its loading reaches only 84.5%.

Therefore, the MCC-2 feeder rather than the transformer is the main limitation for the proposed expansion.

13. Recommended Future Upgrade

Before the future 110 kW motor is installed, the MCC-2 feeder should be reviewed and upgraded.

The upgrade would likely require:

- A higher-capacity MCC-2 feeder
- Re-evaluation of conductor sizing
- Re-evaluation of feeder protection
- Verification of MCC bus capacity
- Updated short-circuit and protection studies
- Updated voltage-drop calculations

The exact upgrade size would require a more detailed code-based design and equipment coordination study.

14. Project Limitations

This project was developed as an educational and portfolio-level industrial power-system design.

Several simplifications were made.

Motor loads were represented as aggregate constant-power loads in the PowerWorld model.

Feeder impedances were simplified rather than developed from final cable length, conductor material, conductor size, and installation method.

Detailed short-circuit calculations and protective-device coordination were outside the scope of this project.

The equipment ratings used in the project should therefore be considered design assumptions rather than construction-ready selections.

A full industrial design would also require detailed application of applicable electrical codes, utility requirements, manufacturer data, grounding studies, arc-flash analysis, and protection coordination.

15. Software Used

EPLAN Education

EPLAN was used for:

- Single-line diagrams
- MCC schematics
- Motor-control circuits
- Protection devices
- Control wiring
- Electrical documentation

PowerWorld Simulator

PowerWorld was used for:

- Power-flow modeling
- Bus-voltage analysis
- Transformer loading analysis
- Feeder loading analysis
- Utility demand analysis
- Operating-case comparison

Google Sheets

Google Sheets was used for:

- Motor calculations
- Transformer sizing
- Feeder sizing
- Load schedules
- Power calculations
- Results comparison
- Graph generation

16. Skills Developed

This project provided practical experience with:

- Industrial electrical distribution
 - Substation design
 - MCC design
 - Three-phase power calculations
 - Transformer sizing
 - Feeder sizing
 - Motor current calculations
 - DOL motor starters
 - Variable frequency drives
 - Motor protection
 - Electrical schematic development
 - Power-flow simulation
 - Bus-voltage analysis
 - Equipment loading analysis
 - Operating-scenario analysis
 - Engineering documentation
-

17. Conclusion

This project demonstrated the complete development of a simplified industrial substation and motor-control distribution system for a food-processing and packaging facility.

The design included a 13.8 kV utility connection, a 750 kVA 13.8 kV/480 V transformer, an 800 A main switchboard, two 300 A motor control centers, an auxiliary distribution section, six industrial motors, and both DOL and VFD motor-control systems.

EPLAN Education was used to develop the single-line and motor-control schematics, while Google Sheets was used for load calculations, transformer selection, feeder sizing, and results analysis.

The completed network was then modeled in PowerWorld Simulator and evaluated under four operating conditions.

The normal and heavy-production cases remained within the selected equipment limits.

Removing the M-202 air compressor significantly reduced MCC-2 loading and overall utility demand.

The future expansion case produced the most important result. Although the transformer remained within its capacity at 84.5% loading, FDR-02 reached 134.1% loading.

The study therefore concludes that the existing transformer has sufficient capacity for the proposed additional motor, but the MCC-2 feeder and associated protection would require an upgrade before the expansion could be implemented.

Overall, the project demonstrated how electrical calculations, schematic design, motor control, and power-flow simulation can be combined to evaluate both current plant operation and future system expansion.